

Graph theory

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1 Basic definition

Definition 1.1: Graph

A graph Γ is a quadruple $(V(\Gamma), E(\Gamma), \phi, \psi, i)$:

- $V(\Gamma)$ and $E(\Gamma)$ are two sets whose elements are called vertices of Γ , edges of Γ , respectively.
- $\phi, \psi : E(\Gamma) \rightarrow V(\Gamma)$ are two mappings determining a choice of starting and end points of each edge e . For each e , $\phi(e)$ is called the **head** of e and $\psi(e)$ is called the **tail** of e .
- An inversion map $i : E(\Gamma) \rightarrow E(\Gamma)$ such that $e \mapsto \bar{e}$ satisfying that

$$\phi(e) = \psi(\bar{e}), \quad \phi(\bar{e}) = \psi(e), \quad e \neq \bar{e}, \quad \bar{\bar{e}} = e$$

An **Orientation** of Γ is a subset $E^+(\Gamma) \subseteq E(\Gamma)$.

Remark The function of ϕ and ψ is to choose the starting point and end point. Once ψ and ϕ exist, a choice is done.

The inversion map hints that the edge set E has a double copy. Unlike the 'prototypical' graph that there are one edge between two vertices, here every two vertices has two edges between them. This can be viewed as a double cover of the prototypical one, and facilitates group actions on them. An inversion just swaps these 'dual' edges. Since each two vertices has two dual edges between them, the orientation is barely a choice of edges.

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2.1 Bipartite graph

2.1.1 Basic definitions

Definition 2.1: Complete bipartite graph

A complete bipartite graph is a bipartite simple graph having a bipartition (A, B) with $|A| = m, |B| = n$ for some m, n , and having the maximum number of edges making it bipartite. This graph is denoted $K_{m,n}$.

2.1.2 Characterisation of bipartite graphs

Theorem 2.1: Characterisation of bipartite graphs

Let Γ be an undirected graph. TFAE:

- (1) Γ is a bipartite graph
- (2) Γ has no odd closed walk
- (3) Γ has no odd cycle
- (4) Γ has no induced odd cycle

Definition 2.2: Edge contraction

For a graph Γ , an **edge contraction** is an operation that removes an edge e from Γ while simultaneously merging the two vertices adjacent to e together.

Definition 2.3: Graph minor

Let Γ, H be two undirected graph. H is a **minor** of $\Gamma \Leftrightarrow H$ can be obtained from Γ by applying a finite sequence of the following operations: (1) edge contraction, (2) edge deleting, (3) removing isolated vertices

3 Matching

3.1 Basic definitions

Definition 3.1: Matching

Let Γ be an undirected graph. A matching M in G is a set $M \subseteq E(\Gamma)$ which are non-loop edges, such that $\forall v \in V(\Gamma)$, v is incident to at most one $e \in M$

Definition 3.2: Vertex cover

Let Γ be an undirected graph. A subset $X \subseteq V(\Gamma)$ is a **vertex cover** of $\Gamma \Leftrightarrow \forall e \in E(\Gamma)$, e has one end in X

Definition 3.3: Matching number

Let Γ be an undirected finite graph. The matching number $\nu(\Gamma)$ is the maximum one of $|M|$ among any matching M of Γ . Equivalently,

$$\nu(\Gamma) := \max\{|M| : M \text{ is a matching of } \Gamma\}$$

Definition 3.4: Vertex cover number

Let Γ be an undirected finite graph. Define

$$\tau(\Gamma) := \min\{|X| : X \text{ is a vertex cover of } \Gamma\}$$

3.2 Matching in bipartite graphs

Theorem 3.1: (Berge)

Theorem 3.2: (König) 1st encounter

Let Γ be a bipartite graph. Then, TFAE:

- (1) Γ has a matching M , with $|M| \geq k$.
- (2) $\nexists X \subseteq V(\Gamma) \ |X| < k \wedge X$ is a vertex cover

Proof:

□

3.3 Perfect matching

Definition 3.5: Perfect matching

Corollary 3.1: Regular bipartite graph $\Rightarrow$ perfect matching

Let Γ be a k -regular bipartite graph. Then, Γ admits a perfect matching.

Proof: (without using Hall's theorem)

□

3.3.1 Characterisation of perfect matching

Hall's theorem gives a characterisation of perfect matching in bipartite graph, while Tutte's theorem gives a characterisation of perfect matching in arbitrary graph.

However, Hall's theorem does not entirely require perfect matching. ¹

Theorem 3.3: (Hall)

Let Γ be a bipartite graph with bipartition (A, B) , TFAE:

- (1) $\exists$ matching M of Γ covering A
- (2) $\forall Y \subseteq A, |Y| \leq |N_\Gamma(Y)|$

Proof: (1) $\Rightarrow$ (2): If $\exists Y \subseteq A, |Y| > |N_\Gamma(Y)|$, then it is impossible to get a matching covering A . Because this matching must assign every $v \in A$ to a $w \in B$ without coincidence. But, at the end there is a v cannot be assigned to a $w \in B$ not used.

(2) $\Rightarrow$ (1) Assume $\forall Y \subseteq A, |Y| \leq |N_\Gamma(Y)|$. $\exists$ matching M of Γ covering $A \Leftrightarrow \exists M, |M| \geq |A| \Leftrightarrow \nu(\Gamma) \geq |A| \Leftrightarrow \tau(\Gamma) \geq |A| \Leftrightarrow \forall$ vertex cover X of $\Gamma, |X| \geq |A|$.

$\forall$ vertex cover X of Γ , consider $A - X$. Notice that $N_\Gamma(A - X) \subseteq B \cap X$. Because $N_\Gamma(A - X) = \bigcup_{v \in A - X} N_\Gamma(v)$. If v is adjacent to some $u \in V(\Gamma)$, $u \in B$ because Γ is bipartite. Also, since X is a vertex cover, in this case the edge corresponding to $\{v, u\}$ already has $v \notin X$, so $u \in X$. Therefore, $u \in B \cap X \Rightarrow N_\Gamma(u) \subseteq B \cap X$. If v does not adjacent to any vertices, $N(v) = \emptyset \subseteq B \cap X$.

Then, by assumption $|A - X| \leq |N_\Gamma(A - X)| \leq |B \cap X|$. Furthermore,

$$|X| = |A \cap X| + |B \cap X| \geq |A \cap X| + |A - X| = |A|$$

□

Remark The reason of considering $A - X$ is that we want the edges to have one end in A and another in B . In the meanwhile, one in X and another is not.

4 Separation

Definition 4.1: Separation

Let Γ be an undirected graph. A separation in Γ is a pair (A, B) , where $A, B \subseteq V(\Gamma)$, such that:

- (1) $A \cup B = V(\Gamma)$
 - (2) $\nexists e \in E(\Gamma), \psi(e) = \{u, v\}$ such that $u \in A \setminus B$ and $v \in B \setminus A$ (one end in $A \setminus B$ and another one in $B \setminus A$)
- In other words, $\forall$ edge with one end in A and another end in B , it has an end in $A \cap B$.

The **order of a separation** (A, B) is $|A \cap B|$

Theorem 4.1

Let $P, Q \subseteq V(\Gamma)$ and $k \in \mathbb{Z}^+$, then **exactly one** of the following holds:

- $\exists$ paths $P_1, \dots, P_k \forall i \neq j \rightarrow \text{Either } V(P_i) \cap V(P_j) = \emptyset$, or P_i has one end in Q and one end in R .
- $\exists$ a separation (A, B) of Γ of order $< k$, $Q \subseteq A$ and $R \subseteq B$.

Theorem 4.2: Menger's theorem

Let Γ be an undirected graph. Let $s, t \in V(\Gamma)$ be distinct and **non-adjacent** and $k \geq 1$ be an integer. Then exactly one of the following holds:

- $\exists$ paths $P_1, \dots, P_k$ from s to t in $\Gamma, \forall i \neq j, V(P_i) - \{s, t\}$ and $V(P_j) - \{s, t\}$ are pairwise disjoint.
- $\exists$ separation (A, B) of $G, s \in A - B$ and $t \in B - A$. (A, B) of order $< k$

¹[Got21]

4.1 Stronger connectedness

Definition 4.2: k -connectedness

Let Γ be an undirected finite graph. Γ is **k -connected** $\Leftrightarrow$

- $|V(\Gamma)| \geq k + 1$
- $\forall X \subseteq V(\Gamma)$ with $|X| \leq k - 1$, $\Gamma_{\setminus X}$ is connected

Corollary 4.1: Existence of k -path in k -connected graph

Let Γ be a k -connected finite undirected graph. Let $s, t \in V(\Gamma)$ with $s \neq t$. Then, $\exists k$ paths $P_1, \dots, P_k$, with the same endpoints (one is s and another is t), and other vertices than s, t are disjoint, i.e. $\forall i \neq j, i \neq j \rightarrow V(P_i - \{s, t\}) \cap V(P_j - \{s, t\}) = \emptyset$

Proof: If s, t are adjacent, then apply Menger's theorem 4.2. It is impossible to get the second case because of k -connectivity of Γ : Suppose Γ admits one separation (A, B) of order $< k$. By 'removing' $A \cap B$ from Γ , the graph $\Gamma_{\setminus A \cap B}$ is connected, meaning $\exists$ an edge with one end in $A - B$ and another end in $B - A$. Contradictions to the fact that (A, B) is a separation.

If s, t are adjacent, then remove all edges between s and t to obtain a graph Γ' . Then, Γ' is at least $k - 1$ -connected. Apply the first part to get $k - 1$ such paths, and add a one-edge path being deleted to get k -paths. $\square$

Corollary 4.2

Let $Q, R \subseteq V(\Gamma)$ and $k \geq 1$ be an integer. Then exactly one of following holds: • $\exists$ paths $P_1, \dots, P_k$ in Γ , $\forall i \neq j, V(P_i)$ and $V(P_j)$ are pairwise disjoint. Each P_i has one end in Q and another end in R . • $\exists$ separation (A, B) of G , of order $< k$. $Q \subseteq A$ and $R \subseteq B$.

Definition 4.3: Cut

A cut in Γ corresponding to X , denoted $\delta(X)$, is a collection of all edges of G with one end in X and another in $V(\Gamma) - X$

5 Directed graphs

In a directed graph, we mean a graph Γ has an orientation $E^+(\Gamma)$. Every discussion should be set up on this orientation.

Definition 5.1: Edges leaving/entering X

Let $X \subseteq V(\Gamma)$. The set of **edges leave** X , denoted $\delta^+(X)$, is defined to be

$$\delta^+(X) := \{e \in E^+(\Gamma) : \phi(e) \in X, \psi(e) \notin X\}$$

Similarly, the set of **edges enter** X is $\delta^-(X) = \{e \in E^+(\Gamma) : \psi(e) \in X, \phi(e) \notin X\}$.

Remark Notice that the $+$ in δ^+ has no relation to the $+$ in E^+ . The set $\delta^+(X)$ simply means all (directed) edges that leave X , meaning it has head in X and tail outside X .

The inversion of edges plays well with this definition:

$$e \in \delta^+(X) \Leftrightarrow \bar{e} \in \delta^-(X)$$

Definition 5.2: Flow

Let Γ be a directed graph and $s, t \in V(\Gamma)$. An $s - t$ **flow** is a function $f : E^+(\Gamma) \rightarrow \mathbb{R}_{\geq 0}$ that $\forall v \in V(\Gamma) - \{s, t\}$,

$$\sum_{e \in \delta^+(\{v\})} f(e) = \sum_{e \in \delta^-(\{v\})} f(e)$$

The **value** of f , $\text{val}(f) := \sum_{e \in \delta^+(\{s\})} f(e) - \sum_{e \in \delta^-(\{s\})} f(e)$.

Remark This definition models a system that has a source s and a sink t , such that for each point v except s, t , the flow entering v has the same amount as the flow leaving v .

In practical flow networks, a vertex is designated as the source is permitted to have both incoming and outgoing edges. Being an origin does not preclude receiving inflows. However, the value of a flow is defined to be the net outflow—the total flow emanating s minus any flow returning to it.

Lemma 5.1: Cut Invariance

Let Γ be a directed graph with an orientation E^+ . Given $X \subseteq V(\Gamma)$ and distinct vertices $s, t \in V(\Gamma)$ with $s \in X, t \notin X$, for any $s - t$ flow $f : E^+(\Gamma) \rightarrow \mathbb{R}_{\geq 0}$ we have

$$\sum_{e \in \delta^+(X)} f(e) - \sum_{e \in \delta^-(X)} f(e) = \text{val}(f)$$

Remark This lemma simply tells one thing: no matter how large a cut is (as long as it does not contain the sink), the net flow leaves this cut (part) is conservative.

'Given $X \subseteq V(\Gamma)$ and distinct vertices $s, t \in V(\Gamma)$ with $s \in X, t \notin X$ ' means that we choose an arbitrary cut that only engulfs s (excluding t), which reveals the conservation.

Proof: From the definition 5.2, $\forall v \in X - \{s\}$, $\sum_{e \in \delta^+(\{v\})} f(e) - \sum_{e \in \delta^-(\{v\})} f(e) = 0$. Hence,

$$\text{val}(f) = \sum_{v \in X} \left(\sum_{e \in \delta^+(\{v\})} f(e) - \sum_{e \in \delta^-(\{v\})} f(e) \right)$$

To swap the sum, the inner index e must be independent of v . So, define characteristic functions as follows

$$\chi_1(e, v) := \begin{cases} 1 & e \in \delta^+(\{v\}) \\ 0 & e \notin \delta^+(\{v\}) \end{cases} \quad \chi_2(e, v) := \begin{cases} 1 & e \in \delta^-(\{v\}) \\ 0 & e \notin \delta^-(\{v\}) \end{cases}$$

Then, val can be rewritten as

$$\text{val}(f) = \sum_{v \in X} \left(\sum_{e \in \delta^+(\{v\})} f(e) - \sum_{e \in \delta^-(\{v\})} f(e) \right) \tag{1}$$

$$= \sum_{v \in X} \left(\sum_{e \in E^+(\Gamma)} f(e) \chi_1(e, v) - \sum_{e \in E^+(\Gamma)} f(e) \chi_2(e, v) \right) \tag{2}$$

$$= \sum_{e \in E^+(\Gamma)} \sum_{v \in X} f(e) (\chi_1(e, v) - \chi_2(e, v)) \tag{3}$$

$$= \sum_{e \in E^+(\Gamma)} f(e) (n_1(e) - n_2(e)) \tag{4}$$

where

$$n_1(e) := \begin{cases} 1 & e \in \delta^+(X) \\ 0 & e \notin \delta^+(X) \end{cases} \quad \chi_2(e, v) := \begin{cases} 1 & e \in \delta^-(X) \\ 0 & e \notin \delta^-(X) \end{cases}$$

So, $\sum_{e \in E^+(\Gamma)} f(e) n_1(e) = \sum_{e \in \delta^+(X)} f(e)$ and $\sum_{e \in E^+(\Gamma)} f(e) n_2(e) = \sum_{e \in \delta^-(X)} f(e)$. Plugging these into (4), we get the desired equality. $\square$

We just modeled a system of flow. In real life, each 'tube' of such a system has an upper bound of flow. The next definition mimics this situation.

Definition 5.3: Capacity

Still Γ is a directed graph with orientation $E^+(\Gamma)$. A function $c : E^+(\Gamma) \rightarrow \mathbb{Z}^+$ is called a capacity function. $\forall e \in E^+(\Gamma)$, $c(e)$ is called the **capacity** of e .
 For an $s - t$ flow f , if $\forall e \in E^+(\Gamma)$, $f(e) \leq c(e)$ for a given capacity function c , then this f is said to be **c -admissible**.

Our objective is to maximize the $s - t$ flow. Intuitively, to maximize the flow leaving s , we can either increase the flow on the outgoing edges, or decrease the flow on the incoming edges. We then apply this logic recursively to adjacent vertices, ensuring we do not backtrack to the previous vertex. This local adjustment can be formalized by the following function:

$$g_{u \rightarrow v}(e) := \begin{cases} +1 & e \in \delta^+(\{v\}) \text{ excluding the edge from } u \text{ to } v, f(e) < c(e) \\ -1 & e \in \delta^-(\{v\}) \text{ excluding the edge from } v \text{ to } u, f(e) > 0 \\ 0 & \text{else} \end{cases}$$

However, a local perturbation at a single vertex is conceptually insufficient because of the flow conservation 5.2. Flow conservation dictates that the total flow entering a vertex v must be equal to the total flow leaving the vertex v . So, if we increase the flow arriving at v , then we create a momentary 'excess' at v . To maintain the balance, this excess must be forwarded immediately to a neighbour vertex.

Consequently, the local adjustment cannot stand alone, it must trigger a chain reaction. If we adjust a vertex incident to v , we are forced to adjust (the flow of) an edge incident to the next vertex. This process propagates the flow adjustment step-by-step until excess reaches the sink s , where the conservation is not required.

Precisely, this valid chain of vertices and local adjustments forms a path, which is defined as an Augmenting Path.

Definition 5.4: Augmenting path

Let Γ be a directed graph with orientation E^+ and flow f . An augmenting path is a simple path $P = (s = v_0, v_1, \dots, v_k = t)$ such that for every consecutive pair of vertices (v_i, v_{i+1}) in the path, the adjustment is possible:

$$g_{v_i \rightarrow v_{i+1}}(e) \neq 0$$

where e is the edge connecting v_i and v_{i+1} .

Remark Why we require that this function being non-zero? This is because if the value is 0, there is no room to adjust the flow on that edge.

Lemma 5.2: Existence of higher capacity flow

Theorem 5.1: Max-flow Min-cut (Ford-Fulkerson)

6 Independent sets and cliques

6.1 Independent sets

Definition 6.1: Independent set

Let Γ be an undirected graph. An **independent set** of Γ is a subset $X \subseteq V(\Gamma)$ such that no two vertices in X are adjacent in Γ .

Definition 6.2: Maximal size of independent set

Let Γ be an undirected graph and X an independent set of Γ . The size of X is the cardinality of X , $|X|$. The maximum size of independent set, denoted $\alpha(\Gamma)$, is defined

$$\alpha(\Gamma) := \max\{|X| : X \text{ is an independent set of } \Gamma\}$$

Lemma 6.1: Independent sets and vertex covering

$$\alpha(\Gamma) + \tau(\Gamma) = |V(\Gamma)|$$

Proof: Let X be an independent set. Then, $V(\Gamma) - X$ is a vertex covering. Since $\forall e \in E(\Gamma)$, it has an end in $V(\Gamma) - X$. Otherwise, e has both ends in X , contradicting the independency of X .

Next, let X be with $|X| = \alpha(X)$, i.e. with the maximal size. Then,

$$|V(\Gamma)| - \alpha(\Gamma) = |V(\Gamma) - \alpha(\Gamma)| = |V(\Gamma) - X| \geq \tau(\Gamma)$$

For the other direction, let Y be a vertex cover. Then, $V(\Gamma) - Y$ is an independent set. Still, let $|Y| = \tau(\Gamma)$, we have

$$|V(\Gamma)| - \tau(\Gamma) = |V(\Gamma) - \tau(\Gamma)| = |V(\Gamma) - Y| \leq \alpha(\Gamma)$$

□

Theorem 6.1: (Gallai)

Let Γ be a **connected simple** graph with $|V(\Gamma)| \geq 2$. Then,

$$\rho(\Gamma) + \nu(\Gamma) = |V(\Gamma)|$$

6.2 Cliques

Definition 6.3: Clique

Let Γ be an undirected graph. A **clique** of Γ is a subset of vertices $C \subseteq V(\Gamma)$ such that $\forall u, v \in V(\Gamma)$ ($u \neq v$), u and v are adjacent. Equivalently, a **clique** of Γ is a subset of vertices $C \subseteq V(\Gamma)$ such that the induced subgraph induced by C in Γ is a complete graph

7 Ramsey theory

In this section, Ramsey theory gives a philosophy: in a large enough system, no matter how chaotic the structure is, there is some subsystem with special, non-chaotic (or ordered) substructures.

Definition 7.1: Ramsey number

$$R(s, t) := \min\{n \in \mathbb{N} : \forall \text{ simple graph } G, |V(G)| = n, \exists \text{ independent set of size } s \vee \text{ a clique of size } t\}$$

This number is saying that: How big must a structure be to ensure that at least one of the pieces has a given interesting property?

7.1 Bounds for Ramsey numbers

What is the concrete number? The answer for the whole picture is not known yet. But, we can still give some bound for it.

Theorem 7.1: Recursive bound for Ramsey number

Let $s, t \geq 1$ be integers. Then, $R(s, t)$ exists. Moreover, $\forall s, t \geq 2$,

$$R(s, t) \leq R(s-1, t) + R(s, t-1)$$

Proof: Induction on $s + t$. Base case: When $s + t = 1$, it is trivial that $R(s, t) = 1$ exists.

Inductive hypothesis: Suppose that when $s + t = k - 1$, $R(s, t)$ exists, and moreover,

$$R(s, t) \leq R(s-1, t) + R(s, t-1)$$

Inductive step: When $s + t = k$, $R(s-1, t)$ and $R(s, t-1)$ exist by inductive hypothesis. It remains to show the inequality. Define $n := R(s-1, t) + R(s, t-1)$, then consider a graph Γ with n vertices. $\forall v \in V(\Gamma)$ Consider two cases:

- $\deg_G(v) \geq R(s, t-1)$ Consider the graph induced by $N(v)$, F . By definition 7.1, F either has an independent set of size s or a clique of size $t-1$.

If it already has an independent set of size s , then this means $\exists$ an independent set of size s in Γ , so $R(s, t) \leq |V(\Gamma)| = n$, done.

If F has a clique of size $t-1$, then $\exists$ a clique C in F of size $t-1$, then adding v and all edges between v and vertices in C , so there is a clique of size t in Γ . So, $R(s, t) \leq |V(\Gamma)| = n$

- $\deg_G(v) \leq R(s, t-1) - 1$ In this case, consider all 'non-neighbours', v has at least (corresponding to $\leq$)

$$n - (R(s, t-1) - 1) - 1 = n - R(s, t-1) = R(s-1, t)$$

non-neighbourhood. Let G be the subgraph induced by all non-neighbours of v .

If G has a clique of size t in G , then there is a clique of size t in Γ , done.

If G has an independent set I in G of size $s-1$, combining v , since v is not adjacent to any vertex in G , then we have an independent set $G \cup \{v\}$ of size s , so by minimality of $R(s, t)$, $R(s, t) \leq |V(\Gamma)| = n$ $\square$

Corollary 7.1: Enhanced recursive bound for Ramsey number

When $R(s, t-1)$ and $R(s-1, t)$ are both even,

$$R(s, t) \leq R(s-1, t) + R(s, t-1) - 1$$

Proof: Denote $n := R(s, t-1) + R(s-1, t)$, Let Γ be a graph with $|V(\Gamma)| = n - 1$. n is even $\Rightarrow n - 1$ is odd. By Handshaking lemma, $\sum_{v \in V(\Gamma)} \deg(v_i) = 2|E(\Gamma)|$, the sum is even. So, there must exist a $v \in V(\Gamma)$ such that $\deg_\Gamma(v)$ is even. Fix that v , denote $M := V(\Gamma) - \{v\} - N(v)$, and $N := N(v)$

Then, notice that $|M| + |N| = |V(\Gamma)| - 1 = n - 2$ is even. Since $|N| = |N(v)| = \deg(v)$ is even, $|M|$ is even. Also, either $|M| \geq R(s-1, t) - 1$ or $|N| \geq R(s, t-1) - 1$. Otherwise, $|M| + |N| < n - 2$.

- If $|M| \geq R(s-1, t) - 1$. In this case, since $|M|$ is even, this can be strengthened to $|M| \geq R(s-1, t)$.

Either the subgraph induced by M , say G_M , has an independent set F of size $s-1$, or G_M has a clique C of size t . In the first case, $F \cup \{v\}$ is an independent set of size s in Γ . In the second case, C is a clique of size t in Γ .

- If $|N| \geq R(s, t-1) - 1$. In this case, since $|N|$ is even, this can be strengthened to $|N| \geq R(s, t-1)$.

Either the subgraph induced by N , say G_N , has an independent set F of size s , or G_N has a clique C of size $t-1$. In the first case, F is an independent set of size s in Γ . In the second case, $C \cup \{v\}$ is a clique of size t in Γ .

In both cases, Γ either has an independent set of size s or a clique of size t . So,

$$R(s, t) \leq |V(\Gamma)| = n - 1 = R(s, t-1) + R(s-1, t) - 1$$

$\square$

Corollary 7.2: Explicit bound for Ramsey number

$\forall s, t \geq 1$,

$$R(s, t) \leq \binom{s+t-2}{s-1}$$

Remark This gives $R(3, 3) = 6$, why? This corollary 7.2 gives $R(3, 3) \leq 6$. Then, consider cycle with 5 vertices, C_5 . C_5 has no independent of size 3 and no clique of size 3. So, $R(3, 3) > 5$.

Proof: Still, induct on $s + t$

Base case: $s = 1$ or $t = 1$, $R(s, t) = 1 = \binom{s-1}{s-1} = \binom{t-1}{0}$

Inductive hypothesis: true for $s + t = k - 1$

Inductive step: when $s + t = k$, by inductive hypothesis,

$$R(s-1, t) \leq \binom{s+t-3}{s-2} \quad R(s, t-1) \leq \binom{s+t-3}{s-1}$$

By theorem 7.1,

$$R(s, t) \leq R(s-1, t) + R(s, t-1) \leq \binom{s+t-3}{s-2} + \binom{s+t-3}{s-1} \stackrel{\dagger}{=} \binom{s+t-2}{s-1}$$

†: Pascal's identity

□

7.2 'Multicoloring' Ramsey theory

In this little section, we extend the classical Ramsey theory to multiple numbers.

In coloring of Ramsey theory, two incident edges are allowed to have the same colour, which is different from the Edge colouring problem. In edge colouring problem, two incident edges are not allowed to be assigned the same colour. To distinguish them, here is the definition:

Definition 7.2: Incident edge colouring

Let Γ be an undirected graph. An incident k -edge colouring is a map

$$c : E(\Gamma) \rightarrow S, \quad |S| = k$$

Definition 7.3: Multicolour Ramsey number

$$R_k(s_1, \dots, s_k),$$

$$R_k(s_1, \dots, s_k) := \min\{n : \forall k\text{-edge colouring of } K_n, \exists i, K_{s_i} \text{ is monochromatic}\}$$

Remark In other words, given the number k , and integers $n_1, \dots, n_k$, $R_k(s_1, \dots, s_k)$ is the minimum integer that for every incident k -edge coloring, there is a complete subgraph on n_i vertices colored i .

Lemma 7.1: Recursive bound for multicolor Ramsey

For all $k \geq 2$,

$$R_k(s_1, \dots, s_k) \leq R_{k-1}(s_1, \dots, s_{k-2}, R(s_{k-1}, s_k))$$

Proof: Consider a complete graph K_n with $n = R_{k-1}(s_1, \dots, s_{k-2}, R(s_{k-1}, s_k))$. Then, we will show K_n has a coloring i on K_{s_i} .

In this case, $\exists 1 \leq i \leq k-2$, such that $\exists$ monochromatically colored K_{s_i} by color i . When it comes to the last number, there exists a monochromatically colored graph $K_{R(s_{k-1}, s_k)}$ by the last color. But, by definition of $R(s_{k-1}, s_k)$, this is equivalent to either the existence of monochromatically colored $K_{s_{k-1}}$ by $k-1$ or the existence of monochromatically colored K_{s_k} by k . By minimality of $R_k(s_1, \dots, s_k)$, $R_k(s_1, \dots, s_k) \leq R_{k-1}(s_1, \dots, s_{k-2}, R(s_{k-1}, s_k))$. □

8 Graph colouring

8.1 Vertex coloring

Definition 8.1: Vertex colouring

Let Γ be an undirected graph and S a set with $|S| = k$ for some $k \in \mathbb{N}$.

A **k -colouring** is a function $\varphi : V(\Gamma) \rightarrow S$ such that $\forall e \in E(\Gamma)$ with $\psi(e) = \{u, v\}$, $\varphi(u) \neq \varphi(v)$

Elements of S are **colours**.

A **colouring set** of (Γ, φ) is a set $\varphi^{-1}(\{s\}) \subseteq V(\Gamma)$ for some $s \in S$

The **colouring class** of (Γ, φ) is the class of all colouring set $\{\varphi^{-1}(\{s\}) : s \in S\}$

Remark If a graph Γ has a loop, then the k -colouring function is not well-defined, since for some loop e , $\psi(e) = \{u\}$, then $\varphi(u) \neq \varphi(u)$, impossible. In this case, we sometimes say $\chi(\Gamma) = \infty$.

Theorem 8.1: The colouring class forms a partition of vertices

Now we introduce vertex colouring problem: Given a graph Γ , what is the minimum k making it k -colourable? This yields the following:

Definition 8.2: Chromatic number

$$\chi(\Gamma) := \min\{k \in \mathbb{N} : \exists k\text{-colouring } \varphi : V(\Gamma) \rightarrow S \text{ for some } S, |S| = k\}$$

- (1) Γ is k -colourable $\Rightarrow \chi(\Gamma) \leq k$
- (2) Γ has one edge $\Rightarrow \chi(\Gamma) \geq 2$
- (3) $\chi(K_n) = n$
- (4) $\chi(E) = 1$, E is an edgeless graph.
- (5) $\chi(C_n) = \begin{cases} 2 & n \text{ is even} \\ 3 & n \text{ is odd} \end{cases}$
- (6) $\chi(P_n) = \begin{cases} 2 & n \geq 2 \\ 1 & n = 1 \end{cases}$
- (7) $\chi(K_{m,n}) = 2$

Corollary 8.1: Chromatic number for bipartite graphs

$$E(\Gamma) \neq \emptyset. \Gamma \text{ is bipartite} \Leftrightarrow \chi(\Gamma) = 2$$

Proof: $\Rightarrow$ Γ is bipartite with bipartition (A, B) , then A could be coloured by a and B could be coloured by b , so we have a 2-colouring function $c : V(\Gamma) \rightarrow \{a, b\}$, since for every edge $e \in E(\Gamma)$, by definition of bipartite graphs, $\psi(e) = \{u, v\}$, with u in A and v in B , so then $c(u) = a \neq b = c(v)$. So, Γ is 2-colourable $\Rightarrow \chi(\Gamma) \leq 2$. Since $E(\Gamma) \neq \emptyset$, so then $\chi(\Gamma) \geq 2$.

$\Leftarrow$ Suppose that $\chi(\Gamma) = 2$, then $\exists$ a 2-colouring $c : V(\Gamma) \rightarrow \{1, 2\}$ Consider

$$A := \{v \in V(\Gamma) : c(v) = 1\} \quad B := \{u \in V(\Gamma) : c(u) = 2\}$$

Claim: (A, B) is a bipartition of Γ . But this is clear: $\forall e \in E(\Gamma)$, $\psi(e) = \{u, v\}$, by definition 8.1, $c(u) \neq c(v)$. So, either $u \in A$ and $v \in B$, or $u \in B$ and $v \in A$. In either case, e has ends in two different sets, so Γ is bipartite. $\square$

8.2 Edge coloring

9 Planar graphs

A planar graph is a graph that can be embedded into a plane such that the edges only intersect at the vertices, i.e. it could be redrawn without any edges crossing.

Now here gives some concrete sense of which graphs are planar and which are not. Consider K_4 . It usually can be drawn with edges intersect. But, it can also be redrawn till there is no edge intersection. However, [Fle17], if you fiddle with K_5 or $K_{3,3}$, there is no way to remove crossings. So, these two graphs are non-planar as we shall see.

Why should we study the planar graphs? Chapter 21 of [Fle17] gives two reasons:

(1) Planar graphs have some interesting mathematical properties: Planar graph can be colored with 4 colors. Facts on planar graphs can show that there are only 5 Platonic solids.

(2) Some practical objects can be modeled as a graph structure involving edge-crossing as a nuisance, for example design problem of subways, circuits and utility lines. Although subway tunnels are crossing in the plan, one subway tunnel is actually buried deeper than another. This burying generates higher expense in construction.

Definition 9.1: Planar graph

A graph is planar if it can be embedded into a genus 0 plane without edge intersections. This embedding without edge-intersection is called a drawing.

Remark As we will prove, K_5 is not a planar graph, it is impossible to embed it into K without edge intersections. But, it can be embedded into a torus without edge intersections.

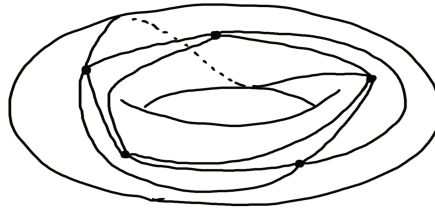


Figure 1: How is K_5 embedded in a torus

Definition 9.2: Faces in graph

Lemma 9.1: Characterisation of cut edge(1st half)

Let G be a graph drawn in the plane and $e \in G$. If the faces on each side of e are the same, then e is a cut edge.

Proof: Suppose that the faces on each side are the same, then $\exists$ an 'edge' in the genus 0 plane embedded avoiding any other edges from one side of e to the other. Closing this curve, by Jordan's theorem, the plane is separated into two parts. Suppose e has two end points u, v , $\phi(e) = \{u, v\}$. Then, u and v are in two different faces. So, u, v are in different components. (different faces imply different parts?) $\square$
The converse is true. But, it needs an extra lemma:

Lemma 9.2: Characterisation of cut edge(2nd half)

Proof: $\square$

Theorem 9.1: Euler's formula

Let Γ be a planar graph. Then,

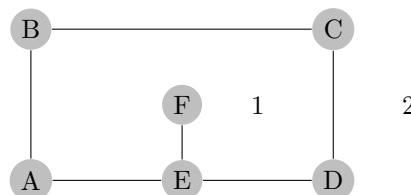
$$|V(\Gamma)| - |E(\Gamma)| + \text{Reg}(\Gamma) = 1 + \text{comp}(\Gamma)$$

Proof: $\square$

Definition 9.3: Degree/Length of a face

Let Γ be a planar graph with a drawing $\tilde{\Gamma}$. The **degree/length** of a face is the minimum number of edges on the boundary of that face forming a closed walk of a face traversing inside the face.

Remark An edge is counted twice if points on both sides of the edge is in the same face.



This is a planar graph. It has two faces. The minimal length of face 1 should be counted inside face 1, and it consists of $AB, BC, CD, DE, EF, FE, EA$ to make the walk closed. So, the length/degree of face 1 is 7. And for face 2, its boundary consists of AB, BC, CD, DE and EA . So, face 2 has length 5.

From this example, some observation suggests that there is a 'face' version of handshaking lemma:

Theorem 9.2: Handshaking lemma for faces

Let Γ be a planar graph. Then, the sum of face degrees is $2|E(\Gamma)|$.

Proof: Let $T(f, e)$ be a function defined as follows:

$$T(f, e) := \begin{cases} 1 & e \text{ is an edge on the boundary of the face } f \\ 0 & e \text{ is an edge on the boundary of the face } f \end{cases}$$

Let F denote the collection of all faces divided by edges in the planarization of Γ . Then,

$$\sum_{f \in F} L(f) = \sum_{f \in F} \sum_{e \in E(\Gamma)} T(f, e) = \sum_{e \in E(\Gamma)} \sum_{f \in F} T(f, e) \stackrel{\dagger}{=} \sum_{e \in E(\Gamma)} 2 = 2|E(\Gamma)|$$

$\dagger$: for a specific e , either there is one face that counts e twice, or this e is shared by two faces. □

Lemma 9.3

9.0.1 Partial characterisation of planar graphs

Lemma 9.4: Edge and vertice numbers of planar graphs

Let Γ be a **connected** simple graph with $|E(\Gamma)| \geq 2$. If Γ is planar, then $|E(\Gamma)| \leq 3|V(\Gamma)| - 6$. Furthermore, if Γ is planar and does not contain a subgraph K_3 , then $|E(\Gamma)| \leq 2|V(\Gamma)| - 4$.

Remark The reason why we assume Γ to be connected is to apply the Euler's formula in 9.1 to get a concrete number.

Proof: We have

$$2|E(\Gamma)| \stackrel{\dagger}{=} \sum_{f \in F} L(f) \stackrel{\ddagger}{\geq} \sum_{f \in F} 3 = 3\text{Reg}(\Gamma) \quad (*)$$

$\dagger$: Comes from theorem 9.2. $\ddagger$: comes from lemma 9.3.

Then, since Γ is connected, apply Euler's formula,

$$2 = |V(\Gamma)| - |E(\Gamma)| + \text{Reg}(\Gamma) \leq |V(\Gamma)| - \frac{1}{3}|E(\Gamma)|$$

So, by arranging it, we get the desired form.

When Γ does not contain any K_3 subgraph, the length/degree of a face is greater than 3, i.e. ≥ 4 . By modifying the equality (*), we have another form. □

Corollary 9.1: Non-planarity of K_5 and $K_{3,3}$

Proof: Apply two inequalities to K_5 and $K_{3,3}$ respectively, done. □

Definition 9.4: Minimal non-planarity

A graph Γ is minimal non-planar $\Leftrightarrow \Gamma$ is non-planar and any proper subgraph of Γ is planar.

K_5 and $K_{3,3}$ are minimal non-planar.

Corollary 9.2: Variation inequalities for a planar graph

Let Γ be a **connected** simple planar graph with $|E(\Gamma)| \geq 2$, and $n_i := |\{v \in V(\Gamma) : \deg_\Gamma(v) = i\}|$. Then,

$$\sum_{v \in V(\Gamma)} (6 - \deg_\Gamma(v)) \geq 12 \quad \text{and} \quad \sum_i (6 - i)n_i \geq 12$$

Proof: From lemma 9.4, we see $12 \geq 6|V(\Gamma)| - 2|E(\Gamma)|$. Since $|V(\Gamma)| = \sum_{v \in V(\Gamma)} 1$ and by Handshaking lemma, $2|E(\Gamma)| = \sum_{v \in V(\Gamma)} \deg_\Gamma(v)$, then we have $12 \geq \sum_{v \in V(\Gamma)} (6 - \deg_\Gamma(v))$. By partitioning $V(\Gamma)$ into sets having different degrees, $V(\Gamma) = \bigsqcup_i V_i$, where $V_i = \{v \in V(\Gamma) : \deg(v) = i\}$. $\sum_{v \in V(\Gamma)} 1 = \sum_{v \in V_1} 1 + \sum_{v \in V_2} 1 + \dots$, we can rewrite $\sum_{v \in V(\Gamma)} (6 - \deg_\Gamma(v))$ as $\sum_i (6 - i)n_i$. $\square$

Corollary 9.3: Degree of a vertex in planar graphs

Let Γ be a simple planar finite graph. Then, $\exists v \in V(\Gamma), \deg_\Gamma(v) < 6$.

Proof: Let Γ be a simple finite graph. If Γ is not connected, consider one of its connected components. So, it suffices to assume that Γ is connected.

When $|V(\Gamma)| = 1$ or 2 , this is clear. When $|V(\Gamma)| > 2$, if $\forall v \in V(\Gamma), \deg_\Gamma(v) \geq 6$, then it must be $|E(\Gamma)| > 2$. Suppose that $\forall v \in V(\Gamma), \deg_\Gamma(v) \geq 6$. Then, $\sum_{v \in V(\Gamma)} (6 - \deg_\Gamma(v)) \leq 0$, contradicting to corollary 9.2. $\square$

9.1 Characterisations of planar, outerplanar**9.1.1 Characterisation of planar graphs**

The two examples of non-planar graphs K_5 and $K_{3,3}$ were not randomly picked. It turns out that any non-planar graph must contain a subgraph that is 'closely related' to one of these two graphs. 'Closed related' actually means 'subdivision' or 'minor' relation.

The lemma below seems unrelated to planarity, but is useful in the proof of Kuratowski's theorem.

Lemma 9.5

Let C be a cycle and $X, Y \subseteq V(C)$ be arbitrary two sets. Then, one of the following is true:

- (1) $\exists u, v \in V(C)$, P, Q are two paths with ends u and v , forming $C \rightarrow X \subseteq V(P) \wedge Y \subseteq V(Q)$.
- (2) $\exists x_1, x_2 \in X, \exists y_1, y_2 \in Y, x_1 \neq x_2 \wedge y_1 \neq y_2 \wedge x_1, x_2, y_1, y_2$ appear on C in the order x_1, y_1, x_2, y_2
- (3) $X = Y$ and $|Y| = 3$

Theorem 9.3: (Kuratowski-Wagner)

Let Γ be a finite graph. Γ is planar $\Leftrightarrow \Gamma$ does not contain K_5 or $K_{3,3}$ as a minor

Proof:

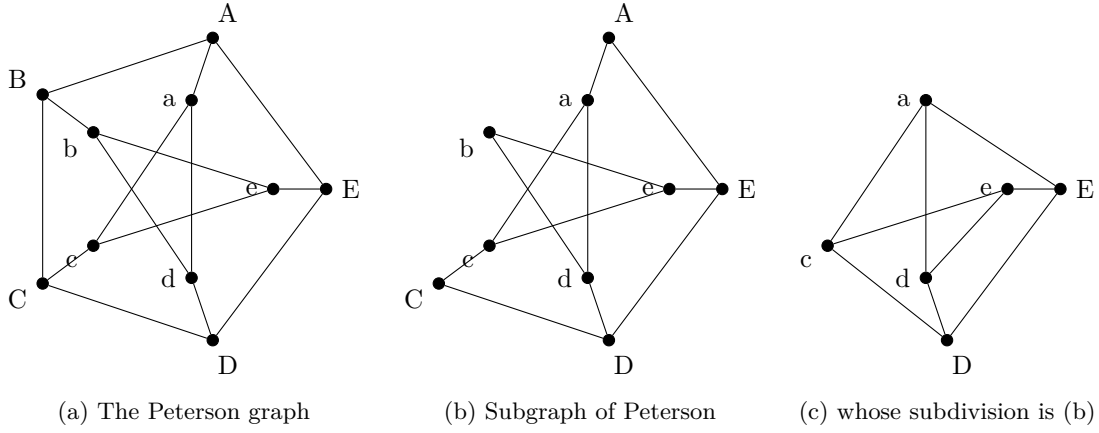
In 1930, Kazimierz Kuratowski proved a theorem, which gives a characterisation of non-planar graphs. $\square$

Theorem 9.4: Kuratowski's theorem

Let Γ be a finite graph. Γ is non-planar $\Leftrightarrow \Gamma$ contains a subgraph that is a subdivision of K_5 or $K_{3,3}$

This is the famous Peterson graph(after a Danish mathematician Julius Peterson). Denote the Peterson graph P . We see that $|V(P)| = 10$ and $|E(P)| = 15$, and it does not have any K_3 subgraph. Since $10 = |V(P)| < 3|E(P)| - 6 = 24$, it satisfies the formula in lemma 9.4. The lemma 9.4 does not give any information on its planarity. But, theorem 9.4 tells us the answer.

In this example, (c) can be subdivided by adding A to the edge aE , C to the edge cD and b to the edge de .



9.1.2 Characterisation of outer planar graph

Definition 9.5: Outerplanar graph

An **outerplanar graph** is a graph that has a planar drawing for which all vertices belong to the outer face of the drawing.

Analogous to Wagner's theorem, outerplanar graphs are characterised by the two forbidden minors K_4 and $K_{2,3}$.

Theorem 9.5: Characterisation of outerplanar graphs

Let Γ be a finite graph. Γ is an outer planar graph $\Leftrightarrow \Gamma$ does not contain a minor of K_4 or $K_{2,3}$

Proof: For any graph Γ , let Γ' be a graph by adding a new vertex v to $V(\Gamma)$ and edges with ends v and vertices ranging over $V(\Gamma)$. Then, we claim that: Γ' is planar $\Leftrightarrow \Gamma$ is outer planar.

If Γ is outer planar, every $w \in V(\Gamma)$ is in the outer/infinite faces. So, there is a drawing connecting v to all $w \in V(\Gamma)$ avoiding edge intersections. Hence, Γ' has a planar drawing and it is planar.

If Γ is not outer planar, then either Γ is not planar, or Γ is planar but with one vertex, say z , inside the face enclosed by some walk for every planar drawing of Γ . For the first case, Γ' is obviously not planar (Minors of planar graphs are planar). For the second case, Let v be in the outer/infinite face. By connecting v to z , there must be an edge-crossing, so then Γ' is also not planar.

So, by Wagner's theorem 9.3 $\dagger$, Γ is outerplanar $\Leftrightarrow \Gamma'$ is planar $\stackrel{\dagger}{\Leftrightarrow} \Gamma'$ does not have K_5 or $K_{3,3}$ minors $\Leftrightarrow \Gamma$ does not have K_4 or $K_{2,3}$ minors (last step by removing that v and edges incident to v). $\square$

9.1.3 Some other characterisations

Theorem 9.6: Characterisation of bipartite planar graph

Let Γ be a graph drawn in the plane. Γ is bipartite $\Leftrightarrow \forall$ face of the planar drawing has even length.

Proof:

Exercise

Exercise 9.1 Let Γ be a 2-connected outerplanar graph.

Exercise 9.2 Let Γ be a 2-connected planar graph. $\forall v \in V(\Gamma)$, define $S(v) := \frac{1}{2} - \frac{1}{\deg_{\Gamma}(v)}$. Then, $\exists$ a face f such that $\sum_{v \in V(\partial f)} S(v) < 1$, where the sum is over all vertices on the boundary of the face f .

Proof: Suppose otherwise, for every face f , $\sum_{v \in V(\partial f)} S(v) \geq 1$. Then, consider the function $T(f, v)$, defined as follows:

$$T(f, v) := \begin{cases} 1 & v \text{ is on the boundary of } f \\ 0 & \text{else} \end{cases}$$

Then, notice that $T(f, v) = \mathbf{1}_v(f) = \mathbf{1}_f(v)$, where $\mathbf{1}$ denotes the indicator function for a single input. Let F denote the collection of all faces divided by planar drawing of Γ . Then,

$$\begin{aligned}
\text{Reg}(\Gamma) &= \sum_{f \in F} 1 \stackrel{[1]}{\leq} \sum_{f \in F} \sum_{e \in V(\partial f)} S(v) = \sum_{f \in F} \sum_{v \in V(\partial f)} S(v) T(f, v) && [1] \text{ by assumption} \\
&= \sum_{f \in F} \sum_{v \in V(\Gamma)} S(v) T(f, v) && \text{property of } T(f, v) \\
&= \sum_{v \in V(\Gamma)} \sum_{f \in F} S(v) T(f, v) && \text{range of } v \text{ is independent of } f \\
&= \sum_{v \in V(\Gamma)} \sum_{f \in F} S(v) \mathbf{1}_v(f) \\
&= \sum_{v \in V(\Gamma)} S(v) \sum_{f \in F} \mathbf{1}_v(f) \\
&= \sum_{v \in V(\Gamma)} S(v) \cdot \deg_{\Gamma}(v) && \star \exists! \deg_{\Gamma}(v) \text{ many faces incident to } v \\
&= \frac{1}{2} \sum_{v \in V(\Gamma)} \deg_{\Gamma}(v) - \sum_{v \in V(\Gamma)} 1 \\
&= |E(\Gamma)| - |V(\Gamma)| && \text{by handshaking lemma}
\end{aligned}$$

$\star$ is the key observation in the proof.

Then, $|V(\Gamma)| - |E(\Gamma)| + \text{Reg}(\Gamma) \leq 0$, contradicting the Euler's formula 9.1. $\square$

9.2 Coloring planar graphs

Planar graphs can be colored much easier than other graphs. Why do we say it is much easier? There is an example: We can show every planar graph can be colored by no more than 6 colors [Fle17]: 5-degenerate

Theorem 9.7: 6-colorability of planar graphs

Let Γ be a simple finite planar graph. Then, $\chi(\Gamma) \leq 6$.

Proof: Induct on $|V(\Gamma)|$:

Base case: if $|V(\Gamma)| = 1$. Then, Γ can be colored by 1 color.

Inductive hypothesis: Suppose that for any graph with $|V(\Gamma)| \leq n$, $\chi(\Gamma) \leq 6$.

Then, consider an arbitrary graph Γ with $|V(\Gamma)| = n + 1$. By corollary 9.3, $\exists$ a vertex v of Γ , $\deg_{\Gamma}(v) < 6$. Consider the graph $\Gamma_{\setminus v}$, which is a graph with n vertices. So, applying the inductive hypothesis, we have $\chi(\Gamma_{\setminus v}) \leq 6$. Since v has at most 5 neighbourhood, its neighbours only use 5 of the available colors. So, there is a spare one assigning to v . $\square$

However, we would like to reduce 6 to some less numbers. It is a fact that every planar graph is 5-colorable. The proof is messy, tedious but not hard.

Theorem 9.8: (Heawood) The five color theorem

Proof: $\square$

When it comes to 4-colors, conjectured by Francis Guthrie in 1852, it is not totally answered (fully solved) until 124 years later, and finally proved by Kenneth Appel and Wolfgang Haken at UIUC. This was reduced and exhaustively checked by computer program. Although not everyone is happy with the involvement of computer, this proof is still regarded legitimate.

Theorem 9.9: (Appel-Haken) The four color theorem

Let Γ be a loopless planar graph. Then, $\chi(\Gamma) \leq 4$

9.2.1 Tools for coloring planar graphs

Definition 9.6: Planar triangulation

Lemma 9.6

Definition 9.7: Planar dual graph

Next is a theorem links the two 'tools' introduced: planar triangulation and planar dual.

Theorem 9.10: Tait's theorem

Let Γ be a planar triangulation and Γ^* be a planar dual of Γ . Then, $\chi(\Gamma) \leq 4 \Leftrightarrow \chi(\Gamma^*) = 3$

9.3 Theories lifting the planarity graphs: embedding graphs into surfaces

As we have established, a planar graph is one that can be embedded into $\mathbb{R}^2$. By the Jordan curve theorem, which states that a closed curve splits the plane into two disjoint regions, an embedding into $\mathbb{R}^2$ is topologically equivalent to an embedding into a sphere $\mathbb{S}^2$ (a closed surface with genus 0).

However, we have already seen that K_5 and $K_{3,3}$ cannot be embedded into a plane (or a sphere) 9.1, but can be embedded into a torus (a surface of genus 1). This observation suggests a natural generalization of planarity: if a graph cannot "fit" on a sphere, can we find a more complex surface that accommodates it?

Let us formalize this hierarchy (of embedding). Up to this point, we have focused on connected, closed, orientable 2-manifolds without boundary. These surfaces are classified entirely by their genus g (the number of "handles"). The focus shifts to Σ , a connected closed orientable 2-manifold without boundary, with g_Σ ranging over all integers.

Since a graph Γ might be embeddable into multiple surfaces of different genera, we define the **genus of the graph**, denoted by γ_Γ , as the minimum genus required for such an embedding:

Definition 9.8: Genus of a graph

Let Σ , Γ be as the above-mentioned and g_Σ be the genus of Σ . Then,

$$\gamma_\Gamma := \min\{g_\Sigma : \exists \text{ embedding } \Gamma \rightarrow \Sigma\}$$

Remark Under this definition,

- Planar graphs are those Γ with $\gamma_\Gamma = 0$.

The Kuratowski's theorem 9.3 tells that every non-planar graph has a minor either as K_5 or as $K_{3,3}$, which can be seen as a topological obstruction for embedding a graph into a plane (equivalently, a sphere $\mathbb{S}^2$). In addition, the definition 9.4 implies that these two graphs are also the minimal one to obstruct the embedding process. This obstruction structure could also be generalized for surfaces M with any genera, as *minimal forbidden minors*.

Definition 9.9: Minimal forbidden minors

The **set of minimal forbidden minors of genus l** is the set of minor

Example • Case $g = 0$ (The Plane): The Kuratowski theorem gives that the class of planar graphs has exactly two minimal forbidden minors, K_5 and $K_{3,3}$.

• Case $g = 1$ (The Torus): What prevents a graph from being embedded into a torus? While the set of MFM for the torus is known to be finite, it is vastly more complex than the planar case, containing over 17,000 graphs. For non-orientable surfaces, the situation can be different. For the Projective Plane, the number of MFM is exactly 35. Since both of them are able to be embedded into a torus, so it is possible to guess that they are the building blocks of any graphs that can be at least embedded into Σ of genus $g_\Sigma = 1$. **where** Since every Σ can be classified by g_Σ , so the study shifts to Σ with all possible genus. This sort of point of view also unified the planar cases, and the universal theory behind it is the Heawood Map Coloring theorem, which swallowed the Four Color theorem 9.9 as a special case of it.

This generalized picture also raise another question: what are the 'minimal' minors that cannot embed into Σ with certain g_Σ ? We can assign a concept for such minors, called the minimal forbidden minors of a graph. Still, the Kuratowski theorem 9.3 tells us planar graphs have only two minimal forbidden minors, K_5 and $K_{3,3}$. What is the answer for a graph that can be embedded at least for a torus? If Σ is a projective plane, then the number of MFM is 35. If Σ is a torus, the number of MFM is greater than 17000. **where**

Given a graph Γ , can we know the minimal genus with which Γ can embed into Σ ? Namely, this is detecting the genus of that graph Γ . This question is NP hard. **where**

Given a graph Γ that can be embedded into Σ with $g_\Sigma := l$, is it possible to embed Γ into a Σ' with $g_{\Sigma'} = l + 1$? The answer is affirmative.

Definition 9.10: Cellular embedding

Theorem 9.11: Duke's theorem

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